

SIMPLE OVERFLOW STOPPER For Water Tanks

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Even though water covers two-thirds of the Earth's surface, less than 1% of it is fresh and easily accessible for human use. Preventing unnecessary wastage of this limited resource is therefore essential. Several methods are currently employed to address this concern, and one effective approach is to use an electronic or electrical system to prevent storage water tanks from overflowing. This also helps conserve electrical energy by ensuring the pump motor runs only long enough to fill the tank with the correct volume of water.

Here we describe an electronic system that uses a dedicated water-level sensor to switch off the pump motor as soon as the tank begins to overflow. Unlike many existing systems, this design does not require any modification to the water tank itself. The sensor is instead mounted at the end of the overflow pipe connected to the tank. The author's prototype, built on a general-purpose PCB, is shown in Fig. 1.

The technique

The system uses magnetic float sensor FS37A, as shown in Fig. 2. This type of liquid-level sensor uses a float containing a permanent magnet. The float moves up and down along a hollow, sealed plastic stem, following the level of the liquid it floats on, and the magnet inside moves with it. The stem is hermetically sealed and houses a reed switch.

The float's initial position relative to the reed switch determines whether the sensor is normally closed (N/C) or normally open (N/O). Fig. 3 shows an



Fig. 1: Author's prototype

N/O-type sensor. In this configuration, the reed switch remains open (off) as long as the float rests on the stopper. As the liquid level rises, the float moves upward, bringing the magnet closer to the reed switch. When the magnet reaches the switch, the switch closes due to the magnetic field. Electric current then flows through the reed switch via connecting wires. A relay-control circuit uses this current to energise or de-energise an electromagnetic relay, which in turn switches the pump motor on or off.

To use this sensor effectively within the system, a sensor assembly is constructed to house the sensor. When the assembly is attached to the overflow pipe of a water tank (see Fig. 4), the overflowing water acts on the float, driving it upward.

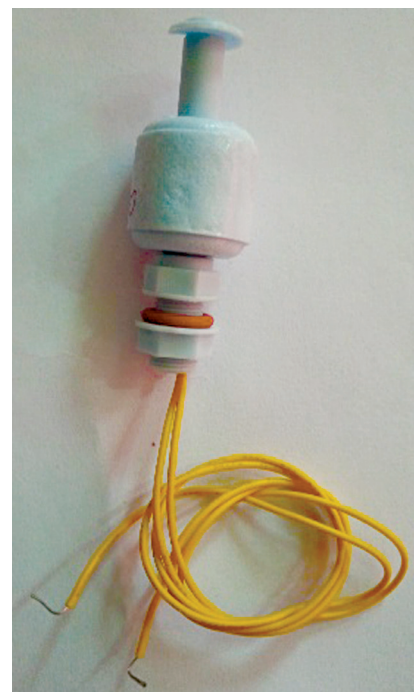


Fig. 2: Float sensor FS37A